APPENDIX A

Trigonometric identities

Formulas for rotation about the principal axes by θ:

$$R_X(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}, \tag{A.1}$$

$$R_Y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}, \tag{A.2}$$

$$R_Z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{A.3}$$

Identities having to do with the periodic nature of sine and cosine:

$$\begin{aligned} \sin\theta &= -\sin(-\theta) = -\cos(\theta + 90^\circ) = \cos(\theta - 90^\circ), \\ \cos\theta &= \cos(-\theta) = \sin(\theta + 90^\circ) = -\sin(\theta - 90^\circ). \end{aligned} \tag{A.4}$$

The sine and cosine for the sum or difference of angles θ_1 and θ_2:

$$\begin{aligned} \cos(\theta_1 + \theta_2) &= c_{12} = c_1 c_2 - s_1 s_2, \\ \sin(\theta_1 + \theta_2) &= s_{12} = c_1 s_2 + s_1 c_2, \\ \cos(\theta_1 - \theta_2) &= c_1 c_2 + s_1 s_2, \\ \sin(\theta_1 - \theta_2) &= s_1 c_2 - c_1 s_2. \end{aligned} \tag{A.5}$$

The sum of the squares of the sine and cosine of the same angle is unity:

$$c^2\theta + s^2\theta = 1. \tag{A.6}$$

If a triangle's angles are labeled a, b, and c, where angle a is opposite side A, and so on, then the "law of cosines" is

$$A^2 = B^2 + C^2 - 2BC\cos a. \tag{A.7}$$

The "tangent of the half angle" substitution:

$$\begin{aligned} u &= \tan\frac{\theta}{2}, \\ \cos\theta &= \frac{1-u^2}{1+u^2}, \\ \sin\theta &= \frac{2u}{1+u^2}. \end{aligned} \tag{A.8}$$

To rotate a vector Q about a unit vector $\hat{K}$ by θ, use **Rodriques's formula**:

$$Q' = Q\cos\theta + \sin\theta(\hat{K} \times Q) + (1 - \cos\theta)(\hat{K} \cdot \hat{Q})\hat{K}. \tag{A.9}$$

See Appendix B for equivalent rotation matrices for the 24 angle-set conventions and Appendix C for some inverse-kinematic identities.

APPENDIX B

The 24 angle-set conventions

The 12 Euler angle sets are given by

$$R_{X'Y'Z'}(\alpha,\beta,\gamma)=\begin{bmatrix} c\beta c\gamma & -c\beta s\gamma & s\beta \\ s\alpha s\beta c\gamma + c\alpha s\gamma & -s\alpha s\beta s\gamma + c\alpha c\gamma & -s\alpha c\beta \\ -c\alpha s\beta c\gamma + s\alpha s\gamma & c\alpha s\beta s\gamma + s\alpha c\gamma & c\alpha c\beta \end{bmatrix},$$

$$R_{X'Z'Y'}(\alpha,\beta,\gamma)=\begin{bmatrix} c\beta c\gamma & -s\beta & c\beta s\gamma \\ c\alpha s\beta c\gamma + s\alpha s\gamma & c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma \\ s\alpha s\beta c\gamma - c\alpha s\gamma & s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma \end{bmatrix},$$

$$R_{Y'X'Z'}(\alpha,\beta,\gamma)=\begin{bmatrix} s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma & s\alpha c\beta \\ c\beta s\gamma & c\beta c\gamma & -s\beta \\ c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma & c\alpha c\beta \end{bmatrix},$$

$$R_{Y'Z'X'}(\alpha,\beta,\gamma)=\begin{bmatrix} c\alpha c\beta & -c\alpha s\beta c\gamma + s\alpha s\gamma & c\alpha s\beta s\gamma + s\alpha c\gamma \\ s\beta & c\beta c\gamma & -c\beta s\gamma \\ -s\alpha c\beta & s\alpha s\beta c\gamma + c\alpha s\gamma & -s\alpha s\beta s\gamma + c\alpha c\gamma \end{bmatrix},$$

$$R_{Z'X'Y'}(\alpha,\beta,\gamma)=\begin{bmatrix} -s\alpha s\beta s\gamma + c\alpha c\gamma & -s\alpha c\beta & s\alpha s\beta c\gamma + c\alpha s\gamma \\ c\alpha s\beta s\gamma + s\alpha c\gamma & c\alpha c\beta & -c\alpha s\beta c\gamma + s\alpha s\gamma \\ -c\beta s\gamma & s\beta & c\beta c\gamma \end{bmatrix},$$

$$R_{Z'Y'X'}(\alpha,\beta,\gamma)=\begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & -s\alpha s\beta s\gamma + c\alpha c\gamma & -s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix},$$

$$R_{X'Y'X'}(\alpha,\beta,\gamma)=\begin{bmatrix} c\beta & s\beta s\gamma & s\beta c\gamma \\ s\alpha s\beta & -s\alpha c\beta s\gamma + c\alpha c\gamma & -s\alpha c\beta c\gamma - c\alpha s\gamma \\ -c\alpha s\beta & c\alpha c\beta s\gamma + s\alpha c\gamma & c\alpha c\beta c\gamma - s\alpha s\gamma \end{bmatrix},$$

$$R_{X'Z'X'}(\alpha,\beta,\gamma)=\begin{bmatrix} c\beta & -s\beta c\gamma & s\beta s\gamma \\ c\alpha s\beta & c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha c\beta s\gamma - s\alpha c\gamma \\ s\alpha s\beta & s\alpha c\beta c\gamma + c\alpha s\gamma & -s\alpha c\beta s\gamma + c\alpha c\gamma \end{bmatrix},$$

$$R_{Y'X'Y'}(\alpha,\beta,\gamma)=\begin{bmatrix} -s\alpha c\beta s\gamma + c\alpha c\gamma & s\alpha s\beta & s\alpha c\beta c\gamma + c\alpha s\gamma \\ s\beta s\gamma & c\beta & -s\beta c\gamma \\ -c\alpha c\beta s\gamma - s\alpha c\gamma & c\alpha s\beta & c\alpha c\beta c\gamma - s\alpha s\gamma \end{bmatrix},$$

$$R_{Y'Z'Y'}(\alpha,\beta,\gamma)=\begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha s\beta & c\alpha c\beta s\gamma + s\alpha c\gamma \\ s\beta c\gamma & c\beta & s\beta s\gamma \\ -s\alpha c\beta c\gamma - c\alpha s\gamma & s\alpha s\beta & -s\alpha c\beta s\gamma + c\alpha c\gamma \end{bmatrix},$$

$$R_{Z'X'Z'}(\alpha,\beta,\gamma)=\begin{bmatrix} -s\alpha c\beta s\gamma + c\alpha c\gamma & -s\alpha c\beta c\gamma - c\alpha s\gamma & s\alpha s\beta \\ c\alpha c\beta s\gamma + s\alpha c\gamma & c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha s\beta \\ s\beta s\gamma & s\beta c\gamma & c\beta \end{bmatrix},$$

$$R_{Z'Y'Z'}(\alpha,\beta,\gamma)=\begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha c\beta s\gamma - s\alpha c\gamma & c\alpha s\beta \\ s\alpha c\beta c\gamma + c\alpha s\gamma & -s\alpha c\beta s\gamma + c\alpha c\gamma & s\alpha s\beta \\ -s\beta c\gamma & s\beta s\gamma & c\beta \end{bmatrix}.$$

The 12 fixed angle sets are given by

$$R_{XYZ}(\gamma,\beta,\alpha)=\begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix},$$

$$R_{XZY}(\gamma,\beta,\alpha)=\begin{bmatrix} c\alpha c\beta & -c\alpha s\beta c\gamma + s\alpha s\gamma & c\alpha s\beta s\gamma + s\alpha c\gamma \\ s\beta & c\beta c\gamma & -c\beta s\gamma \\ -s\alpha c\beta & s\alpha s\beta c\gamma + c\alpha s\gamma & -s\alpha s\beta s\gamma + c\alpha c\gamma \end{bmatrix},$$

$$R_{YXZ}(\gamma,\beta,\alpha)=\begin{bmatrix} -s\alpha s\beta s\gamma + c\alpha c\gamma & -s\alpha c\beta & s\alpha s\beta c\gamma + c\alpha s\gamma \\ c\alpha s\beta s\gamma + s\alpha c\gamma & c\alpha c\beta & -c\alpha s\beta c\gamma + s\alpha s\gamma \\ -c\beta s\gamma & s\beta & c\beta c\gamma \end{bmatrix},$$

$$R_{YZX}(\gamma,\beta,\alpha)=\begin{bmatrix} c\beta c\gamma & -s\beta & c\beta s\gamma \\ c\alpha s\beta c\gamma + s\alpha s\gamma & c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma \\ s\alpha s\beta c\gamma - c\alpha s\gamma & s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma \end{bmatrix},$$

$$R_{ZXY}(\gamma,\beta,\alpha)=\begin{bmatrix} s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma & s\alpha c\beta \\ c\beta s\gamma & c\beta c\gamma & -s\beta \\ c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma & c\alpha c\beta \end{bmatrix},$$

$$R_{ZYX}(\gamma,\beta,\alpha)=\begin{bmatrix} c\beta c\gamma & -c\beta s\gamma & s\beta \\ s\alpha s\beta c\gamma + c\alpha s\gamma & -s\alpha s\beta s\gamma + c\alpha c\gamma & -s\alpha c\beta \\ -c\alpha s\beta c\gamma + s\alpha s\gamma & c\alpha s\beta s\gamma + s\alpha c\gamma & c\alpha c\beta \end{bmatrix},$$

$$R_{XYX}(\gamma,\beta,\alpha)=\begin{bmatrix} c\beta & s\beta s\gamma & s\beta c\gamma \\ s\alpha s\beta & -s\alpha c\beta s\gamma + c\alpha c\gamma & -s\alpha c\beta c\gamma - c\alpha s\gamma \\ -c\alpha s\beta & c\alpha c\beta s\gamma + s\alpha c\gamma & c\alpha c\beta c\gamma - s\alpha s\gamma \end{bmatrix},$$

$$R_{XZX}(\gamma,\beta,\alpha)=\begin{bmatrix} c\beta & -s\beta c\gamma & s\beta s\gamma \\ c\alpha s\beta & c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha c\beta s\gamma - s\alpha c\gamma \\ s\alpha s\beta & s\alpha c\beta c\gamma + c\alpha s\gamma & -s\alpha c\beta s\gamma + c\alpha c\gamma \end{bmatrix},$$

$$R_{YXY}(\gamma,\beta,\alpha)=\begin{bmatrix} -s\alpha c\beta s\gamma + c\alpha c\gamma & s\alpha s\beta & s\alpha c\beta c\gamma + c\alpha s\gamma \\ s\beta s\gamma & c\beta & -s\beta c\gamma \\ -c\alpha c\beta s\gamma - s\alpha c\gamma & c\alpha s\beta & c\alpha c\beta c\gamma - s\alpha s\gamma \end{bmatrix},$$

$$R_{YZY}(\gamma,\beta,\alpha) = \begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha s\beta & c\alpha c\beta s\gamma + s\alpha c\gamma \\ s\beta c\gamma & c\beta & s\beta s\gamma \\ -s\alpha c\beta c\gamma - c\alpha s\gamma & s\alpha s\beta & -s\alpha c\beta s\gamma + c\alpha c\gamma \end{bmatrix},$$

$$R_{ZXZ}(\gamma,\beta,\alpha) = \begin{bmatrix} -s\alpha c\beta s\gamma + c\alpha c\gamma & -s\alpha c\beta c\gamma - c\alpha s\gamma & s\alpha s\beta \\ c\alpha c\beta s\gamma + s\alpha c\gamma & c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha s\beta \\ s\beta s\gamma & s\beta c\gamma & c\beta \end{bmatrix},$$

$$R_{ZYZ}(\gamma,\beta,\alpha) = \begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha c\beta s\gamma - s\alpha c\gamma & c\alpha s\beta \\ s\alpha c\beta c\gamma + c\alpha s\gamma & -s\alpha c\beta s\gamma + c\alpha c\gamma & s\alpha s\beta \\ -s\beta c\gamma & s\beta s\gamma & c\beta \end{bmatrix}.$$

APPENDIX C

Some inverse-kinematic formulas

The single equation

$$\sin\theta = a \tag{C.1}$$

has two solutions, given by

$$\theta = \pm\text{Atan2}(\sqrt{1-a^2}, a). \tag{C.2}$$

Likewise, given

$$\cos\theta = b, \tag{C.3}$$

there are two solutions:

$$\theta = \text{Atan2}(b, \pm\sqrt{1-b^2}). \tag{C.4}$$

If both (C.1) and (C.3) are given, then there is a unique solution given by

$$\theta = \text{Atan2}(a, b). \tag{C.5}$$

The transcendental equation

$$a\cos\theta + b\sin\theta = 0 \tag{C.6}$$

has the two solutions

$$\theta = \text{Atan2}(a, -b) \tag{C.7}$$

and

$$\theta = \text{Atan2}(-a, b). \tag{C.8}$$

The equation

$$a\cos\theta + b\sin\theta = c, \tag{C.9}$$

which we solved in Section 4.5 with the tangent-of-the-half-angle substitutions, is also solved by

$$\theta = \text{Atan2}(b, a) \pm \text{Atan2}(\sqrt{a^2+b^2-c^2}, c). \tag{C.10}$$

The set of equations

$$\begin{aligned} a\cos\theta - b\sin\theta &= c, \\ a\sin\theta + b\cos\theta &= d, \end{aligned} \tag{C.11}$$

which was solved in Section 4.4, also is solved by

$$\theta = \text{Atan2}(ad - bc,\ ac + bd). \tag{C.12}$$

Solutions to selected exercises

CHAPTER 2 SPATIAL DESCRIPTIONS AND TRANSFORMATIONS EXERCISES

2.1)

$$
\begin{aligned}
R &= ROT(\hat{x}, \phi)ROT(\hat{z}, \theta) \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 0 & C\phi & -S\phi \\ 0 & S\phi & C\phi \end{bmatrix} \begin{bmatrix} C\theta & -S\theta & 0 \\ S\theta & C\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} C\theta & -S\theta & 0 \\ C\phi S\theta & C\phi C\theta & -S\phi \\ S\phi S\theta & S\phi C\theta & C\phi \end{bmatrix}
\end{aligned}
$$

2.12) Velocity is a "free vector" and will be affected only by rotation, not by translation:

$$
{}^{A}V = {}^{A}_{B}R\,{}^{B}V = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix}
$$

$$
{}^{A}V = \begin{bmatrix} -1.34 & 22.32 & 30.0 \end{bmatrix}^{T}
$$

2.27)

$$
{}^{A}_{B}T = \begin{bmatrix} -1 & 0 & 0 & 3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

2.33)

$$
{}^{B}_{C}T = \begin{bmatrix} -0.866 & -0.5 & 0 & 3 \\ 0 & 0 & +1 & 0 \\ -0.5 & 0.866 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

CHAPTER 3 MANIPULATOR KINEMATICS EXERCISES

3.1)

α_{i-1}	a_{i-1}	d_i
0	0	0
0	L_1	0
0	L_2	0

$$
{}^{0}_{1}T = \begin{bmatrix} C_1 & -S_1 & 0 & 0 \\ S_1 & C_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{1}_{2}T = \begin{bmatrix} C_2 & -S_2 & 0 & L_1 \\ S_2 & C_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^{2}_{3}T = \begin{bmatrix} C_3 & -S_3 & 0 & L_2 \\ S_3 & C_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{0}_{3}T = {}^{0}_{1}T\,{}^{1}_{2}T\,{}^{2}_{3}T = \begin{bmatrix} C_{123} & -S_{123} & 0 & L_1C_1 + L_2C_{12} \\ S_{123} & C_{123} & 0 & L_1S_1 + L_2S_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

where

$$
C_{123} = \cos(\theta_1 + \theta_2 + \theta_3)
$$
$$
S_{123} = \sin(\theta_1 + \theta_2 + \theta_3)
$$

3.8) When $\{G\} = \{T\}$, we have

$$
{}^{B}_{W}T\,{}^{W}_{T}T = {}^{B}_{S}T\,{}^{S}_{G}T
$$

So

$$
{}^{W}_{T}T = {}^{B}_{W}T^{-1}\,{}^{B}_{S}T\,{}^{S}_{G}T
$$

4.14) No. Pieper's method gives the closed-form solution for any 3-DOF manipulator. (See his thesis for all the cases.)

4.18) 2

4.22) 1

CHAPTER 5 JACOBIANS: VELOCITIES AND STATIC FORCES EXERCISES

5.1) The Jacobian in frame $\{0\}$ is

$$
{}^{\circ}J(\theta) = \begin{bmatrix} -L_1S_1 - L_2S_{12} & -L_2S_{12} \\ L_1C_1 + L_2C_{12} & L_2C_{12} \end{bmatrix}
$$

$$
\begin{aligned}
\mathrm{DET}({}^{\circ}J(\theta)) &= -(L_2C_{12})(L_1S_1 + L_2S_{12}) + (L_2S_{12})(L_1C_1 + L_2C_{12}) \\
&= -L_1L_2S_1C_{12} - L_2^2S_{12}C_{12} + L_1L_2C_1S_{12} + L_2^2S_{12}C_{12} \\
&= L_1L_2C_1S_{12} - L_1L_2S_1C_{12} = L_1L_2(C_1S_{12} - S_1C_{12}) \\
&= L_1L_2S_2
\end{aligned}
$$

$\therefore$ The same result as when you start with ${}^{3}J(\theta)$, namely, the singular configurations are $\theta_2 = 0°$ or $180°$.

5.8) The Jacobian of this 2-link is

$$^3J(\theta) = \begin{bmatrix} L_1S_2 & 0 \\ L_1C_2 + L_2 & L_2 \end{bmatrix}$$

An isotropic point exists if

$$^3J = \begin{bmatrix} L_2 & 0 \\ 0 & L_2 \end{bmatrix}$$

So

$$L_1S_2 = L_2$$
$$L_1C_2 + L_2 = 0$$

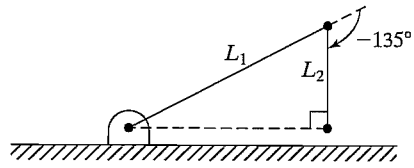
Now, $S_2^2 + C_2^2 = 1$, so $\left(\frac{L_2}{L_1}\right)^2 + \left(\frac{-L_2}{L_1}\right)^2 = 1.$

or $L_1^2 = 2L_2^2 \rightarrow L_1 = \sqrt{2}L_2$.

Under this condition, $S_2 = \frac{1}{\sqrt{2}} = \pm.707$.

and $C_2 = -.707$.

$\therefore$ An isotropic point exists if $L_1 = \sqrt{2}L_2$, and in that case it exists when $\theta_2 = \pm 135°$.



In this configuration, the manipulator looks momentarily like a Cartesian manipulator.

5.13)

$$\tau = {}^\circ J^T(\theta)^\circ F$$

$$\tau = \begin{bmatrix} -L_1S_1 - L_2S_{12} & L_1C_1 + L_2C_{12} \\ -L_2S_{12} & L_2C_{12} \end{bmatrix} \begin{bmatrix} 10 \\ 0 \end{bmatrix}$$

$$\tau_1 = {}^-10S_1L_1 - 10L_2S_{12}$$

$$\tau_2 = -10L_2S_{12}$$

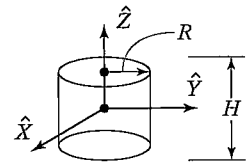
CHAPTER 6 MANIPULATOR DYNAMICS EXERCISES

6.1) Use (6.17), but written in polar form, because that is easier. For example, for I_{zz},

$$I_{zz} = \int_{-H/2}^{H/2} \int_0^{2\pi} \int_0^R (x^2 + y^2) p\, r\, dr\, d\theta\, dz$$

$$x = R\cos\theta, \quad y = R\sin\theta, \quad x^2 + y^2 = R^2(r^2)$$

$$I_{zz} = \int_{-H/2}^{H/2} \int_0^{2\pi} \int_0^R pr^3 dr\, d\theta dz$$



$$I_{zz} = \frac{\pi}{2} R^4 H p, \text{VOLUME} = \pi r^2 H$$

$$\therefore \quad \text{Mass} = M = p\pi r^2 H \quad \therefore \quad \boxed{I_{zz} = \tfrac{1}{2} M R^2}$$

Similarly (only harder) is

$$\boxed{I_{xx} = I_{yy} = \tfrac{1}{4} M R^2 + \tfrac{1}{12} M H^2}$$

From symmetry (or integration),

$$\boxed{I_{xy} = I_{xz} = I_{yz} = 0}$$

$\therefore$

$$c_I = \begin{bmatrix} \frac{1}{4}MR^2 + \frac{1}{12}MH^2 & 0 & 0 \\ 0 & \frac{1}{4}MR^2 + \frac{1}{12}MH^2 & 0 \\ 0 & 0 & \frac{1}{2}MR^2 \end{bmatrix}$$

6.12) $\theta_1(t) = Bt + ct^2$, so

$$\dot{\theta}_1 = B + 2ct, \ddot{\theta} = 2c$$

so

$$^1\dot{\omega}_1 = \ddot{\theta}_1 \hat{z}_1 = 2c\hat{z}_1 = \begin{bmatrix} 0 \\ 0 \\ 2c \end{bmatrix}$$

$$^1\dot{v}_{c1} = \begin{bmatrix} 0 \\ 0 \\ 2c \end{bmatrix} \otimes \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} \otimes \left(\begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} \otimes \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \right)$$

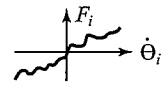
$$= \begin{bmatrix} 0 \\ 4c \\ 0 \end{bmatrix} + \begin{bmatrix} -2\dot{\theta}_1^2 \\ 0 \\ 0 \end{bmatrix}$$

$$^1\dot{v}_{c1} = \begin{bmatrix} -2(B + 2ct)^2 \\ 4c \\ 0 \end{bmatrix}$$

6.18) Any reasonable $F(\theta, \dot{\theta})$ probably has the property that the friction force (or torque) on joint i depends only on the velocity of joint i, i.e.,

$$F(\theta, \dot{\theta}) = [f_1(\theta, \dot{\theta}_1) \quad F_2(\theta, \dot{\theta}_2) \quad \ldots . F_N(\theta, \dot{\theta}_N)]^T$$

Also, each $f_i(\theta, \dot{\theta}_i)$ should be "passive"; i.e., the function should lie in the first & third quadrants.



** Solution written by candlelight in aftermath of 7.0 earthquake, Oct. 17, 1989!

CHAPTER 7 TRAJECTORY GENERATION EXERCISES

7.1) Three cubics are required to connect a start point, two via points, and a goal point—that is, three for each joint, for a total of 18 cubics. Each cubic has four coefficients, so 72 coefficients are stored.

7.17) By differentiation,

$$\dot{\theta}(t) = 180t - 180t^2$$
$$\ddot{\theta}(t) = 180 - 360t$$

Then, evaluating at $t = 0$ and $t = 1$, we have

$$\theta(0) = 10 \quad \dot{\theta}(0) = 0 \quad \ddot{\theta}(0) = 180$$
$$\theta(1) = 40 \quad \dot{\theta}(1) = 0 \quad \ddot{\theta}(1) = -180$$

8.3) Using (8.1), we have

$$L = \sum_{i=1}^{3}(a_{i-1} + d_i) = (0+0) + (0+0) + (0 + (U - L)) = U - L$$

$$W = \frac{4}{3}\pi U^3 - \frac{4}{3}\pi L^3 = \frac{4}{3}\pi(U^3 - L^3) \left\{ \begin{array}{c} \text{a} \\ \text{"hollow"} \\ \text{sphere} \end{array} \right.$$

$$\therefore \quad Q_L = \frac{U - L}{\sqrt[3]{\frac{4}{3}\pi(U^3 - L^3)}}$$

8.6) From (8.14),

$$\frac{1}{K_{TOTAL}} = \frac{1}{1000} + \frac{1}{300} = 4.333 \times 10^{-3}$$

$$\therefore \quad \boxed{K_{TOTAL} = 230.77 \ \frac{\text{NTM}}{\text{RAD}}}$$

8.16) From (8.15),

$$K = \frac{G\pi d^4}{32L} = \frac{(0.33 \times 7.5 \times 10^{10})(\pi)(0.001)^4}{(32)(0.40)} = \boxed{0.006135 \ \frac{\text{NTM}}{\text{RAD}}}$$

This is very flimsy, because the diameter is 1 mm!

9.2) From (9.5),

$$s_1 = -\frac{6}{2 \times 2} + \frac{\sqrt{36 - 4 \times 2 \times 4}}{2 \times 2} = -1.5 + 0.5 = -1.0$$
$$s_2 = -1.5 - 0.5 = -2.0$$

$\therefore$
$quadx(t) = c_1 e^{-t} + c_2 e^{-2t}$ and $\dot{x}(t) = -c_1 e^{-t} - 2c_2 e^{-2t}$

$$Att = 0 \quad x(0) = 1 = c_1 + c_2 \tag{1}$$
$$\dot{x}(0) = 0 = -c_1 - 2c_2 \tag{2}$$

Adding (1) and (2) gives

$$1 = -c_2$$

so $c_2 = -1$ and $c_1 = 2$.

$$\therefore \boxed{x(t) = 2e^{-t} - e^{-2t}}$$

9.10) Using (8.24) and assuming aluminum yields

$$K = \frac{(0.333)(2 \times 10^{11})(0.05^4 - 0.04^4)}{(4)(0.50)} = 123{,}000.0$$

Using info from Figure 9.13, the equivalent mass is $(0.23)(5) = 1.15$ kg, so

$$W_{\text{res}} = \sqrt{k/m} = \sqrt{\frac{123{,}000.0}{1.15}} \cong \boxed{327.04 \ \frac{\text{rad}}{\text{sec}}}$$

This is very high—so the designer is probably wrong in thinking that this link vibrating represents the lowest unmodeled resonance!

9.13) As in problem 9.12, the effective stiffness is $K = 32000$. Now, the effective inertia is $I = 1 + (0.1)(64) = 7.4$.

$$\therefore \quad W_{\text{res}} = \sqrt{\frac{32000}{7.4}} \cong 65.76 \frac{\text{rad}}{\text{sec}} \cong \boxed{10.47 \text{ Hz}}$$

CHAPTER 10 NONLINEAR CONTROL OF MANIPULATORS EXERCISES

10.2) Let $\tau = \alpha\tau' + \beta$

$$\alpha = 2 \quad \beta = 5\theta\dot{\theta} - 13\dot{\theta}^3 + 5$$

and $\tau' = \ddot{\theta}_D + K_v\dot{e} + K_p e$
where $e = \theta_D - \theta$
and

$$K_p = 10$$
$$K_v = 2\sqrt{10}$$

10.10) Let $f = \alpha f' + \beta$
with $\alpha = 2$, $\beta = 5x\dot{x} - 12$
and $f' = \ddot{X}_D + k_v\dot{e} + k_p e, \qquad e = X_D - X$
$k_p = 20$, $k_v = 2\sqrt{20}$.

CHAPTER 11 FORCE CONTROL OF MANIPULATORS EXERCISES

11.2) The artificial constraints for the task in question would be

$V_z = -a_1$	$F_x = 0$ $F_y = 0$ $N_x = 0$ $N_y = 0$ $N_z = 0$

where a_1 is the speed of insertion.

11.4) Use (5.105) with frames $\{A\}$ and $\{B\}$ reversed. First, find ${}^B_AT$, so invert ${}^A_BT$:

$$ {}^B_AT = \begin{bmatrix} 0.866 & 0.5 & 0 & -8.66 \\ -0.5 & 0.866 & 0 & 5.0 \\ 0 & 0 & 1 & -5.0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Now,

$$ {}^BF = {}^B_AR\,{}^AF = \begin{bmatrix} 1 & 1.73 & -3 \end{bmatrix}^T $$

$$ {}^BN = {}^BP_{AORG} \otimes {}^BF + {}^B_AR\,{}^AN = \begin{bmatrix} -6.3 & -30.9 & -15.8 \end{bmatrix}^T $$

$$ \therefore \quad {}^BF = \begin{bmatrix} 1.0 & 1.73 & -3 & -6.3 & -30.9 & -15.8 \end{bmatrix}^T $$

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ENAE 692 Introduction to Space Robotics *
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(January 16, 2006)

Abstract

This document contains errata for the 3rd Edition of John J. Craig's introductory robotics text compiled by myself and the students of ENAE 692 "Introduction to Space Robotics" at the University of Maryland. The 3rd Edition contains many carry over errors from the 2nd Edition and many additional errors perhaps due to being retypeset (please see the companion document for the 2nd Edition). Credit goes to Mads Paulin of the Maersk Mc-Kinney Moller Institute for Production Technology, University of Southern Denmark for many of the errors cited in this document. Sadly, there appear to be many more where those came from, so this will likely be work in progress for the foreseeable future.

Chapter 2

Page 29 In Example 2.2, ${}^{B}P$ in the third line should be

$$ {}^{B}P = [3.0\ 7.0\ 0.0]^T $$

Page 40 The skew symmetric matrix S in equation (2.57) should be

$$ S = \begin{bmatrix} 0 & -s_z & s_y \\ s_z & 0 & -s_x \\ -s_y & s_x & 0 \end{bmatrix} $$

Page 45 In the section entitled "Z-Y-Z Euler angles," the last phrase should be "about $\hat{Z}_B$ by an angle γ."

Page 46 The rotation matrix $R_X(\theta)$ in equation (2.77) should be

$$ R_X(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & cos\theta & -sin\theta \\ 0 & sin\theta & cos\theta \end{bmatrix} $$

*Class Instructor: Craig R. Carignan, Adjunct Faculty, Dept. of Aerospace Engineering, craigc@ssl.umd.edu; thanks to Mads Paulin of the University of Southern Denmark who published the original list of errata online at: *http://www.mip.sdu.dk/~paulin/Courses/ROB00/Craig_3rd_ed_errata.pdf*

Page 58 In Exercise 2.20, the $\hat{Q}$ in Rodriques' Formula should be Q.

Page 59 In Figure 2.26, the top of the wedge is supposed to be a 3-4-5 triangle so 36.9° should be $tan^{-1}(\frac{3}{4})$. [Note that in the 2nd Edition, it's supposed to be a 30-60-90 deg triangle!]

Chapter 3

Page 67 The D-H parameters just above the horizontal line before Example 3.2 should be (a_i, α_i, d_i).

Page 83 The expression for r_{11} in equation (3.14) should be

$$r_{11} = c_1[c_{23}(c_4c_5c_6 - s_4s_6) - s_{23}s_5c_6] + s_1(s_4c_5c_6 + c_4s_6)$$

Chapter 4

Page 109 The last axis of the manipulator in Figure 4.7 should be $\hat{X}_3$ instead of $\hat{Z}_3$.

Page 112 The angle β in Figure 4.8 should start from the $\hat{X}_0$ axis, not the dotted line.

Chapter 5

Page 137 The ${}^U_TR$ in the last line of Example 5.1 should not be there.

Page 139 The vector ${}^AQ$ in Figure 5.3 should be ${}^BQ$.

Page 142 The skew symmetric matrix S in equation (5.25) should be

$$S = \begin{bmatrix} 0 & -\Omega_z & \Omega_y \\ \Omega_z & 0 & -\Omega_x \\ -\Omega_y & \Omega_x & 0 \end{bmatrix}$$

Page 157 In equation (5.94), $\delta\theta$ should be $\delta\Theta$.

Page 158 The matrix operator $P\times$ in equation (5.101) should be

$$P\times = \begin{bmatrix} 0 & -p_z & p_y \\ p_z & 0 & -p_x \\ -p_y & p_x & 0 \end{bmatrix}$$

Page 159 Equation (5.107), as well as the preceding paragraph, are incorrect. The velocity and force transformations in equations (5.103) and (5.105) are not transposes of each other. While the submatrix blocks transpose positions, the individual submatrices do not transpose, e.g. ${}^BR_A \neq {}^BR_A{}^T$ in general.

Page 160 The following exercises have citation errors:

Exercise 5.1 should say, "Repeat Example 5.4 using the Jacobian in (5.67) written in frame {0}."

Exercise 5.2 should cite Exercise 3.3 of Chapter 3.

Exercise 5.3 should cite Exercise 3.3 of Chapter 3.

Exercise 5.10 should cite Example 5.3.

Chapter 6

Page 169-170 The $\omega's$ in Example 6.1 should all be replaced with $w's$.

Page 171 The $\omega's$ in Example 6.2 should all be replaced with $w's$.

Page 173 The following equations have errors:

Equation (6.33) should be ${}^i\dot{\omega}_i$ on the right side.

Equation (6.34) should be ${}^i\dot{\omega}_i$ for first term in brackets on the right side.

Equation (6.36) should be ${}^i\omega_i \times {}^iP_{C_i}$ in the parentheses on the right side.

Page 184-185 The I_{zz2}'s in equations (6.80), (6.81), (6.85), (6.88), and (6.89) should all be replaced with I_{yy2}'s. [Note that the out-of-plane axis for link 2 is the y_2-axis, not the z_2-axis.]

Page 185 The (2,1) element of the vector in equation (6.85) should be $m_2\dot{d}_2$.

Page 195 The center of mass inertia for link 1 should be labeled as ${}^{C_1}I_1$ not C_1I in exercises 6.8, 6.9, 6.11, and 6.12. Likewise, C_3I should be ${}^{C_3}I_3$ in Exercise 6.9.

Chapter 7

Page 214 The t_{12} in the denominator of equation (7.25) should be t_{d12}.

Page 236 The workspace for the SCARA manipulator in Figure 8.5 should be a cylinder centered about the first joint axis, not the second.

Chapter 8

Page 249 The definition of wall thickness in the sentence preceding Example 8.4 should be $(w_o - w_i)/2$. The wall thickness in Example 8.4 must be 0.5 cm to be consistent with the numerator in equation (8.25).

Page 250 In equation (8.25), the denominator should be $4(0.5)^3$ and the right side is 1.48×10^6. This results in a deflection of 6.8×10^{-6} on the right side of equation (8.26) and 0.00068 cm, not 0.027 cm as stated afterwards. Therefore, the 0.027 gets replaced by 0.001 in equation (8.30) resulting in a deflection of 1.631 cm, which is approximately that due to the input shaft deflection alone.

Chapter 9

Page 269 Example 9.2 contains the following errors:

Equation (9.23) should be $-\frac{1}{2}c_1 + \frac{\sqrt{3}}{2}c_2 = 0$.

Immediately following equation (9.23), the root should be $c_2 = -\frac{\sqrt{3}}{3}$.

The argument of the cosine in equation (9.25) should be $(\frac{\sqrt{3}}{2}t - 120°)$.

Page 276 The right side of equation (9.53) should be f_{dist}/m.

Page 287 In Exercise 9.10, the wall thickness should be 0.5 cm instead of 1 cm. Also, no material is specified, but the solutions assume that aluminum is used.

Chapter 10

Page 315 In Exercise 10.14, K_P is assumed to be diagonal.

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